

GREAT ORTHOGONALITY THEOREM

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The Great (or Grand) Orthogonality Theorem, or GOT, is a relation between the elements of an irreducible unitary matrix representation of a group. For a d -dimensional representation $D(g)$ of a group G with elements g , the theorem states that

$$\sum_{g \in G} D^\dagger(g)^i_j D(g)^k_\ell = \frac{N(G)}{d} \delta^i_\ell \delta^k_j \quad (1)$$

where $N(G)$ is the number of elements in G . The theorem is a relation between specific elements of the matrices $D(g)$ and $D^\dagger(g)$ rather than the matrices as a whole, so the LHS of this equation is a single number, not a matrix.

Zee gives a proof in Chapter II.2, but it's worth going through it to clarify a few small points.

We define a matrix A as

$$A \equiv \sum_{g'} D^\dagger(g') X D(g') \quad (2)$$

where X is an arbitrary $d \times d$ matrix. We now consider:

$$D^\dagger(g) A D(g) = D^\dagger(g) \sum_{g'} D^\dagger(g') X D(g') D(g) \quad (3)$$

$$= \sum_{g'} D^\dagger(g) D^\dagger(g') X D(g') D(g) \quad (4)$$

Note that the sum is over g' and that g is some fixed group element. From the multiplication theorem and the closure axiom, $D(g') D(g) = D(g'g)$, and summing this over g' is equivalent to summing $D(g)$ over g . Also, since $[D(g') D(g)]^\dagger = D^\dagger(g) D^\dagger(g')$, we have

$$D^\dagger(g) A D(g) = \sum_{g'} D^\dagger(g'g) X D(g'g) \quad (5)$$

$$= \sum_g D^\dagger(g) X D(g) \quad (6)$$

$$= A \quad (7)$$

Multiplying both sides by $D(g)$ and using the fact that since D is unitary, $D^\dagger = D^{-1}$, we have

$$D(g) D^\dagger(g) A D(g) = D(g) A \quad (8)$$

$$A D(g) = D(g) A \quad (9)$$

This relation is true for all elements $g \in G$, so we can invoke Schur's lemma, which states that any matrix A that commutes with all irreducible representation matrices of a group G is a multiple of the identity, so we have

$$A = \lambda I \quad (10)$$

We can now take the trace of 2 and use the fact that the trace of a product of matrices is cyclic, so that

$$\text{tr} [D^\dagger(g) X D(g)] = \text{tr} [D(g) D^\dagger(g) X] = \text{tr} X \quad (11)$$

Therefore, since X doesn't depend on g , and the trace of I is just d , the dimension:

$$\text{tr} A = \lambda d = \sum_g \text{tr} X = N(G) \text{tr} X \quad (12)$$

so

$$\lambda = \frac{N(G)}{d} \text{tr} X \quad (13)$$

We therefore have

$$A = \frac{N(G)}{d} (\text{tr} X) I \quad (14)$$

We can now specify X to be a $d \times d$ matrix with all elements equal to zero except for element X^j_k which is equal to 1. The trace of X will therefore be non-zero only if X^j_k is a diagonal element, so

$$\text{tr} X = \delta^j_k \quad (15)$$

and

$$A = \frac{N(G)}{d} \delta^j_k I \quad (16)$$

Inserting this X back into 2, we consider

$$A^i_\ell \equiv \sum_g \left[D^\dagger(g) X D(g) \right]^i_\ell \quad (17)$$

For the matrix X with a 1 in row j and column k , the product $D^\dagger(g) X$ copies column j of $D^\dagger(g)$ into column k of $D^\dagger(g) X$, with the remaining columns being zero. If we then calculate $[D^\dagger(g) X] D(g)$ the resulting matrix consists of the non-zero column from $D^\dagger(g) X$ multiplied by row k of $D(g)$. For example, if we're dealing with general 3×3 matrices and $X^2_1 = 1$, then we have for some general matrices A and B :

$$AXB = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \quad (18)$$

$$= \begin{bmatrix} a_{12} & 0 & 0 \\ a_{22} & 0 & 0 \\ a_{32} & 0 & 0 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \quad (19)$$

$$= \begin{bmatrix} a_{12}b_{11} & a_{12}b_{12} & a_{12}b_{13} \\ a_{22}b_{11} & a_{22}b_{12} & a_{22}b_{13} \\ a_{32}b_{11} & a_{32}b_{12} & a_{32}b_{13} \end{bmatrix} \quad (20)$$

Then we have

$$[AXB]^i_\ell = A^i_2 B^1_\ell \quad (21)$$

In general for any size matrices and a general X^j_k

$$[AXB]^i_\ell = A^i_j B^k_\ell \quad (22)$$

Thus from 17 we have

$$A^i_\ell \equiv \sum_g \left[D^\dagger(g) \right]^i_j D(g)^k_\ell \quad (23)$$

From 16 we have

$$\sum_g \left[D^\dagger(g) \right]^i_j D(g)^k_\ell = \frac{N(G)}{d} \delta^i_\ell \delta^j_k \quad (24)$$

which proves 1.

Example 1. For the unitary 2-dim representation of the permutation group S_3 , we have the matrices

$$\begin{aligned} U_I &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; & U_{12} &= \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}; & U_{13} &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}; \\ U_{23} &= \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}; & U_{123} &= \begin{bmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix}; & U_{132} &= \begin{bmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned} \quad (25)$$

This group has $N(G) = 6$ and the representation has $d = 2$, so 1 gives us

$$\sum_{g \in G} U^\dagger(g)^i_j U(g)^k_\ell = 3\delta^i_\ell \delta^k_j \quad (26)$$

Using Maple to do the matrix calculations, we can verify a few cases. If we define $P^{ik}_{j\ell} = \sum_{g \in G} U^\dagger(g)^i_j U(g)^k_\ell$ we find

$$\begin{aligned} P^{11}_{11} &= 3 \\ P^{12}_{12} &= 0 \\ P^{12}_{21} &= 3 \\ P^{12}_{22} &= 0 \end{aligned} \quad (27)$$

and so on. Thus the theorem works in this case.

Example 2. The S_3 group also has a *non-unitary* 2-dim representation

$$\begin{aligned} M_I &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; & M_{I2} &= \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}; & M_{I3} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}; \\ M_{23} &= \begin{bmatrix} -1 & 0 \\ -1 & 1 \end{bmatrix}; & M_{I23} &= \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}; & M_{I32} &= \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix} \end{aligned} \quad (28)$$

If we try calculating 1 for these matrices, we get, with $Q^{ij}_{k\ell} = \sum_{g \in G} M^\dagger(g)^i_j M(g)^k_\ell$

$$\begin{aligned} Q^{11}_{11} &= 4 \\ Q^{12}_{12} &= -1 \\ Q^{12}_{21} &= 4 \\ Q^{12}_{22} &= -2 \end{aligned} \quad (29)$$

and so on. Thus 1 is *not* true if the representation is not unitary.

There is a corollary to 1 which Zee proves in detail in his Appendix 1 of Chapter II.2, so I won't go through the proof here. It states that if we have two non-equivalent irreducible representations r and s , then

$$\sum_{g \in G} D^{(r)\dagger}(g)^i_j D^{(s)}(g)^k_\ell = 0 \quad (30)$$

That is, different representations are always orthogonal. Note that the representations r and s need not have the same dimension, so r could be a 1-dimensional representation and s a 3-dimensional representation.

With this corollary, we can state a more general form of 1 as

$$\boxed{\sum_{g \in G} D^{(r)\dagger}(g)^i_j D^{(s)}(g)^k_\ell = \frac{N(G)}{d} \delta^{rs} \delta^i_\ell \delta^k_j} \quad (31)$$

The symbol δ^{rs} is a sort of Kronecker delta that is zero if the representation r is not equivalent to s , even though r and s aren't actually numbers; they are symbols that define a particular irreducible representation.

PINGBACKS

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